Objective: Simplifying a given function using K-map and implement it using logic gate.

Problem 1:

problem specification: Design and implement a circuit using gates that would take 2-bit numbers as input and produces the sum of input and output.

Truth table:

a_1	a_0	b_1	b_0	c_1	s_1	s_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

K-map for S_0:

$a_1a_0 \backslash b_1b_0$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

$$S_0 = a_0 b_0' + b_0 a_0' \Rightarrow S_0 = a_0 \oplus b_0$$

K-map for S_1:

$a_1a_0 \backslash b_1b_0$	00	01	11	10
00	0	0	1	1
01	0	1	0	1
11	1	0	1	0
10	1	1	0	0

$$S_1 = a_1 b_1' a_0' + a_1 b_0' b_1' + a_1' b_1 b_0' + b_1 a_0' a_1' + a_0 a_1 b_1 b_0 + a_1' b_1' a_0 b_0$$

$$= (a_0' + b_0')(a_1 b_1' + a_1' b_1) + a_0 b_0 (a_1 b_1 + a_1' b_1')$$

$$= (a_0 b_0)' (a_1 \oplus b_1) + (a_0 b_0)(a_1 \oplus b_1)'$$

$$= (a_1 \oplus b_1) \oplus (a_0 b_0)$$

K-map for C_1:

$a_1 a_0$ \ $b_1 b_0$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	1

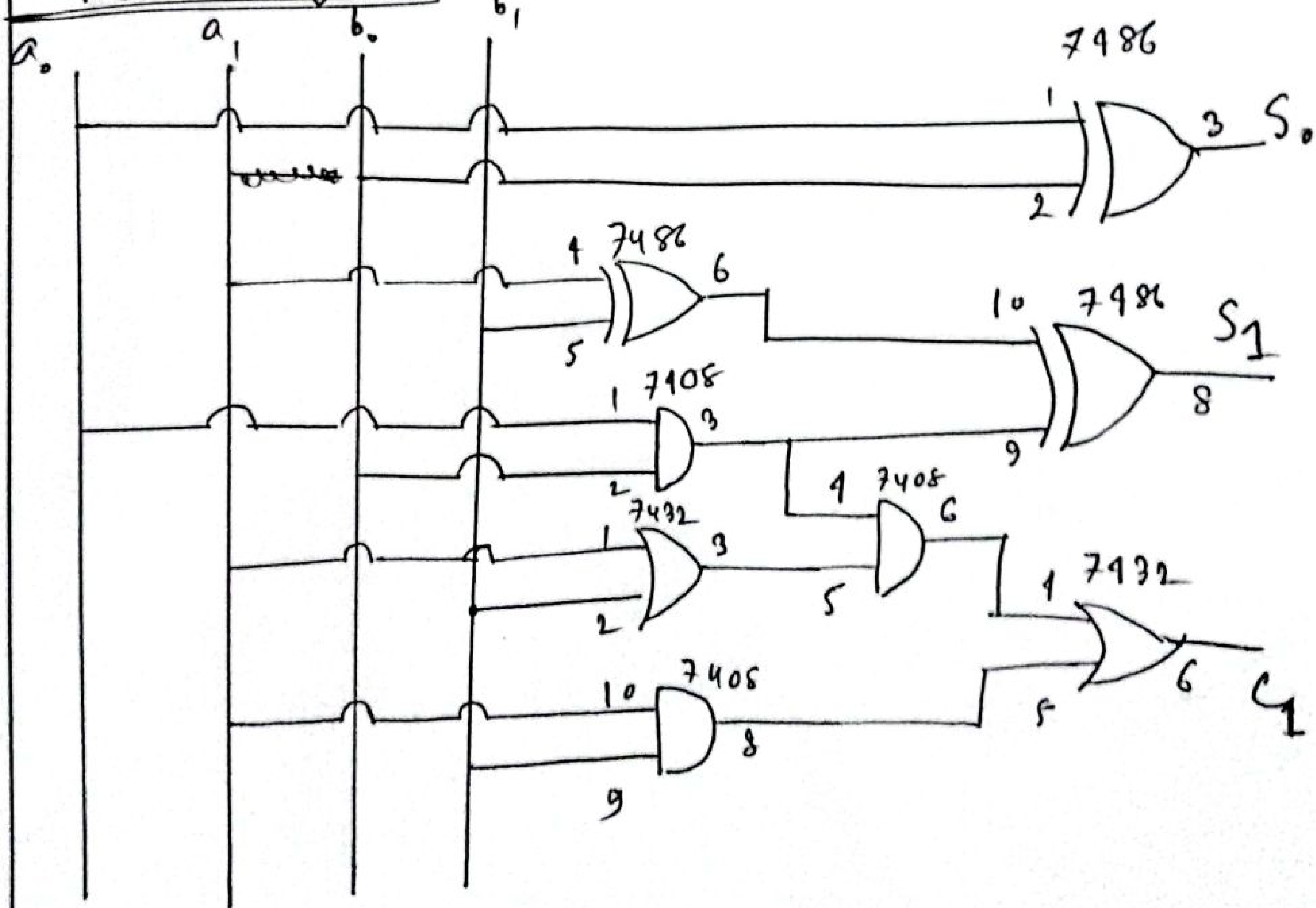
$$C_1 = a_1 b_1 + a_1 b_0 a_0 + a_0 b_0 b_1$$

$$= a_1 b_1 + a_0 b_0 (a_1 + b_1)$$

Required Instruments:

1. Trainer board
2. IC-extractor
3. IC (Quad 2 input AND) 74LS08
4. IC (Quad 2 input OR) 74LS32
5. IC (Quad 2 input X-OR) 7486
6. Wires
7. Handbook.

Circuit diagram:



problem 2:

Problem specification:

We need to simplify and implement the boolean functions with gates:

$$f(V,W,X,Y,Z) = \sum(1,4,5,13,20,21,22,28) + d(6,9,11,12,14,29,30)$$

Truth table:

V	W	X	Y	Z	F
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	0
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	0	1	1
0	0	1	1	0	X
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	0	1	X
0	1	0	1	0	0
0	1	0	1	1	X
0	1	1	0	0	X
0	1	1	0	1	1
0	1	1	1	0	X
0	1	1	1	1	0
1	0	0	0	0	0
1	0	0	0	1	0
1	0	0	1	0	0

V	W	X	Y	Z	F
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	0	1	0
1	1	0	1	0	0
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	0	1	X
1	1	1	1	0	X
1	1	1	1	1	0

K-map for the circuit:

$\overline{V}$

WX \ YZ				
	0	1	0	0
	1	1	0	X
	X	1	0	X
	0	X	X	0

V=0

V

WX \ YZ	00	01	11	10
00	0	0	0	0
01	1	1	0	1
11	1	X	0	X
10	0	0	0	0

V=1

$$F = V'Y'Z + XY' + XZ'$$
$$= V'Y'Z + X(Y' + Z')$$

Required Instruments:

1. Trainer board
2. IC-Extractor
3. IC (Hex-Inverter) - 7404
4. IC (Quad 2 input AND) - 7408
5. IC (Quad 2 Input OR) - 7432
6. Wires
7. Handbook.

Circuit Diagram: